CH Varun

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github.com/TheCoder30ec4

Education

B.Tech in Department of Computer and Communication Engineering — Aug 2021 – present
Amrita Vishwa Vidyapeetham, Chennai, Current Aggregate:7.3CGPA — Chennai, India

Skills

Languages: C| Python| Java| Java Scripts
Frame Works: My SQL| MangoDB| React JS| Next JS| Tailwind CSS| Node JS| AWS Cloud
Machine Learing: Data Science| CNNs| Computer Vision| NLP| Reinforment learning| LSTM| GAN| LLMs
Network Communication: : Experience with TCP/IP | Network Sensors | Cisco Packet Tracer

Projects

AI-Powered Real-Time Plant Disease Detection and Prediction System

- Curated and labeled a diverse dataset of 10,000+ plant disease images. This enriched training data enabled the model to accurately identify a wider spectrum of pathologies compared to previous approaches.
- Achieved 98% disease detection accuracy using CNNs and advanced deep learning methods, surpassing industry standards and revolutionizing diagnosis compared to traditional methods.
- Constructed a user-friendly Django web application for real-time disease detection, empowering over 500 farmers and professionals with instant identification for informed decisions and timely interventions.

Context-Aware Robot Path Prediction for Enhanced Agricultural Efficiency

- Utilized computer vision techniques to extract key environmental features such as obstacles and terrain variations, incorporating them into a robust path prediction algorithm with an accuracy improvement of over 85%.
- Engineered a system capable of predicting robot paths with 92% accuracy, significantly improving field efficiency and autonomous operation capabilities.
- Led a two-person team to a top-three finish at Vit Agrithon hackathon, showcasing our project's agricultural impact. 70% of potential users found our solution valuable, emphasizing its real-world potential.

Sensor-Fusion based Elderly Fall Detection System with Enhanced Prediction and Monitoring

- Revamped a Generative Adversarial Network (GAN) system to predict body joint values from a single sensor, slashing hardware requirements by 90%. This led to substantial cost savings and simplified the system's complexity
- Implemented TCP/IP communication to enable wireless functionality for the BNO055 IMU sensor, enhancing user mobility and comfort by eliminating cable constraints while ensuring reliable data transmission.
- Managed a team of 3, strategically delegating tasks, providing supportive guidance, and fostering a collaborative and motivated environment. This ensured project success and team member growth.

Professional Experience

Full Stack Developer intern — May 2024 – present
Deep Dive Labs, Signapore

- Created a site using NextJs for compliance officer regulation management, integrating AI like ChatGPT and Amazon Bedrock which can analyze over 10,000 regulations.
- Deployed 5 API endpoints, 8 Lambda functions, and DynamoDB for database management. Managed access and security with IAM roles, supporting 1,000+ user sessions daily.
- Handled CRUD operations on a database with 50,000+ entries, using AI-genrated summaries to cut manual processing time by 70%.

Government Internship — Jul 2022 – Aug 2022
Water and Land Management Training and Research Institute(WALAMTARI)

- Developed technology addressing a critical environmental challenge (water scarcity) through data-driven decision-making and resource optimization upto 90%.
- Employed sensor fusion techniques to combine data from the 10 sensors for comprehensive environmental monitoring. Leveraged GSM mobile communication for wireless data transmission from 10 sensors.
- Designed and deployed a user-friendly mobile app for farmers to monitor and manage water usage, which empowered 50+ farmers with real-time water usage optimization capabilities.

Live-in-Labs — Jun 2024 – Jun 2024

- Lead a team of 5 in Bhadrapura, conducting research by interviewing 100+ villagers, demonstrating strong leadership and research skills.
- Organized awareness sessions in schools, engagin 65+ students and teachers which helped in Identifing 15 key issues imapacting daily life.
- Held brainstorming sessions with 50+ villagers, developing solutions for waste management and solar power to address power cuts.

Certificates

- **AMCAT Data Processing Specialist**
- **Machine Learing and Data Science in Udemy**
- **Soft Skills IBM Certification**
- **C-programming in Udemy**